PONDICHERRY UNIVERSITY
RAMANUJAN SCHOOL OF MATHEMATICAL SCIENCES
DEPARTMENT OF STATISTICS

M.Sc. QUANTITATIVE FINANCE
MSQF 533: Financial Risk Management
LECTURE NOTES ON

Financial Risk Management

PREPARED BY
NAME: N Rohit Vedhanandh
REG. NO.: 25MSQUFPY0002

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MSQF 533: FINANCIAL RISK MANAGEMENT CREDITS: 4

Course Objectives:

- *This paper focuses the basic concept of risk management and expose various types of risk faced.*
- *It helps to take positions for investing and trading in options and future market.*

Course Outcomes (COs):

CO1: Analyse how futures and forward markets operate and be able to calculate its theoretical prices and values

CO2: Analyse the sources of financial risk and the importance of implementing effective financial risk management

CO3: Evaluate hedging strategies using forwards, futures, options and swaps

CO4: To identify financial risks in currencies, interest rates, commodities and shares

CO5: To evaluate the outcomes of strategies to manage Risk with the help of case Studies

Unit I: Introduction to Risk Management

Introduction to risk management — Sources of risk — Risks of commodity investment — Risks of equity investment — Risks of fixed income investment — Risks of currency investment — Market risk measurement.

Unit II: VAR Methods and Hedging

An overview of VAR — Definition, downside VAR — VAR methods: VAR local and full valuation, delta normal methods, historical Simulation, Monte Carlo simulation — VAR applications — Stress testing — Hedging: linear risk, optimal hedging, hedge ratio as regression coefficient, non-linear risk hedging.

Unit III: Credit Risk Management

Introduction to credit risk — Measuring credit risk — Credit exposure — Types of credit derivatives — Credit default swap — Pricing and hedging credit derivatives — Credit risk models — Basel accord — The Basel market risk charges, Settlement risk.

Unit IV: Management of Financial Risk

Measuring & Managing Interest Rate Risks: Discounted Cash Flow, Time Value of Money, Yield to Maturity, Fixed Term Annuities & Present Value of Basis Point — Measuring & Managing Market Risk: Risk/Return Trade-off, Systematic Risk, Idiosyncratic Risk, Modern Portfolio Theory & Risk Adjusted Return on Capital (RAROC) — Measuring & Managing Liquidity Risk: Forecasting cash requirements from Cash Flow Analysis, Working Capital Management.

Unit V: Operation & Integrated Risk Management

Measuring & Managing Operational Risk: Brainstorming, SWOT, Root Cause Analysis & Assumption Analysis — Measuring & Managing Credit & Counterparty Risk: Syndication of Credit, Netting of Payments (Cross Collateralization), Third Party Guarantees (Banks) & Credit Derivatives — Measuring Systemic & Sovereign Risk: Assessing Debt/GDP, Debt/Revenue, Debt Servicing, BoP Position and FOREX Reserves & Demographic Profile.

Books for Reference:

1. Crouhy, M., Galai, D., & Mark, R. (2023). *The Essentials of Risk Management* (3rd ed.). McGraw-Hill.
2. Hopkin, P., & Thompson, C. (2022). *Fundamentals of Risk Management: Understanding, Evaluating and Implementing Effective Enterprise Risk Management* (6th ed.). Kogan Page Ltd.
3. Lee, H. (2021). *Risk Management: Fundamentals, Theory, and Practice in Asia*. Springer Singapore.
4. Hunziker, S. (2021). *Enterprise Risk Management: Modern Approaches to Balancing Risk and Reward*. Springer Gabler.
5. Indian Institute of Banking & Finance (IIBF). (2023). *Risk Management*. Macmillan.
6. Jorion, P. (2003). *Financial Risk Management Handbook*. Wiley.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	XXX	XX	XX	X
CO2	XXX	XX	XX	XXX	XX
CO3	XXX	XXX	XXX	XX	XX
CO4	XX	XX	XX	XXX	X
CO5	XX	XX	XXX	XXX	XX

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Chapter 1

Introduction to Risk Management

Lecture
14/08/2026

Financial risk management lies at the core of quantitative finance and modern capital allocation. Investors, financial institutions, and multinational corporations operate in an environment characterized by rapid price fluctuations, interconnected global liquidity, and regulatory transformations. This chapter establishes the fundamental concepts, taxonomy, governance frameworks, and asset-class specific risks that form the bedrock of the subject.

1.1 Foundations of Risk and Uncertainty

Definition 1.1.1: Financial Risk and Knightian Uncertainty

Financial Risk describes a decision environment in which future outcomes are uncertain, but the possible payoff states and their corresponding probability distribution can be mathematically modeled or estimated from historical data and market prices.
In contrast, **Knightian Uncertainty** (formulated by Frank Knight in 1921) refers to situations where future events or outcomes cannot be defined, enumerated, or assigned objective or subjective probability distributions.

In financial markets, risk has a dual character:

- **Downside Exposure:** The potential for financial loss, bankruptcy, capital impairment, or underperformance relative to a target benchmark.
- **Upside Volatility:** The potential for unexpected surplus returns, representing the compensation (risk premium) demanded by rational, risk-averse investors for bearing uncertainty.

Statistical Measures of Return and Dispersion

Let P_t denote the price of a financial asset at time t. The simple discrete return R_t and continuously compounded log-return r_t over period Δt are defined as:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}, \qquad r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(1 + R_t) \tag{1.1}$$

The expected return μ, variance σ^2, and standard deviation (volatility) σ are given by:

$$\mu = \mathbb{E}[r_t], \qquad \sigma^2 = \mathrm{Var}(r_t) = \mathbb{E}\left[(r_t - \mu)^2\right], \qquad \sigma = \sqrt{\mathrm{Var}(r_t)} \tag{1.2}$$

When returns are measured at daily intervals, the annualized volatility σ_{ann} assuming $N = 252$ trading days is:

$$\sigma_{\text{ann}} = \sigma_{\text{daily}} \times \sqrt{252} \tag{1.3}$$

Financial asset returns routinely exhibit non-normal empirical characteristics, specifically **excess kurtosis** (fat tails or leptokurtosis) and **negative skewness** (asymmetric crash risk). Relying exclusively on variance (σ^2) severely underestimates the likelihood of extreme catastrophic market events.

1.2 Taxonomy and Classification of Financial Risks

Total risk in any financial portfolio can be categorized through two fundamental analytical lenses: by its diversifiability within capital markets, and by its underlying structural origin.

1.2.1 Systematic vs Unsystematic Risk

The total variance of an asset's return σ_i^2 under the Single Index Model (Capital Asset Pricing Model framework) is decomposed into two orthogonal components:

$$\sigma_i^2 = \underbrace{\beta_i^2 \sigma_m^2}_{\text{Systematic Risk}} + \underbrace{\sigma_{\epsilon_i}^2}_{\text{Unsystematic (Idiosyncratic) Risk}} \tag{1.4}$$

Systematic Risk (Market-Wide)

Nature: Pervasive, non-diversifiable risk stemming from macro-environmental forces.
Key Drivers:

- Inflationary spikes & monetary policy
- GDP contractions & business cycles
- Geopolitical conflicts & war
- Global liquidity shocks & FX volatility

Mitigation: Cannot be eliminated by diversification; managed via index hedging, asset allocation, and macro overlay strategies.

Unsystematic Risk (Specific)

Nature: Idiosyncratic, diversifiable risk unique to a specific firm or narrow sector.
Key Drivers:

- Corporate governance failures / fraud
- Operational disruption & labour strikes
- Regulatory penalties & patent expirations
- Unexpected earnings downgrades

Mitigation: Substantially eradicated in large portfolios ($N \geq 30$) through uncorrelated cross-sector diversification.

Figure 1.1: Systematic vs Unsystematic Risk Decomposition (Symmetrical Dual-Pillar Architecture).

1.2.2 Comprehensive Risk Taxonomy

Modern financial risk management categorizes exposures into five major interrelated domains:

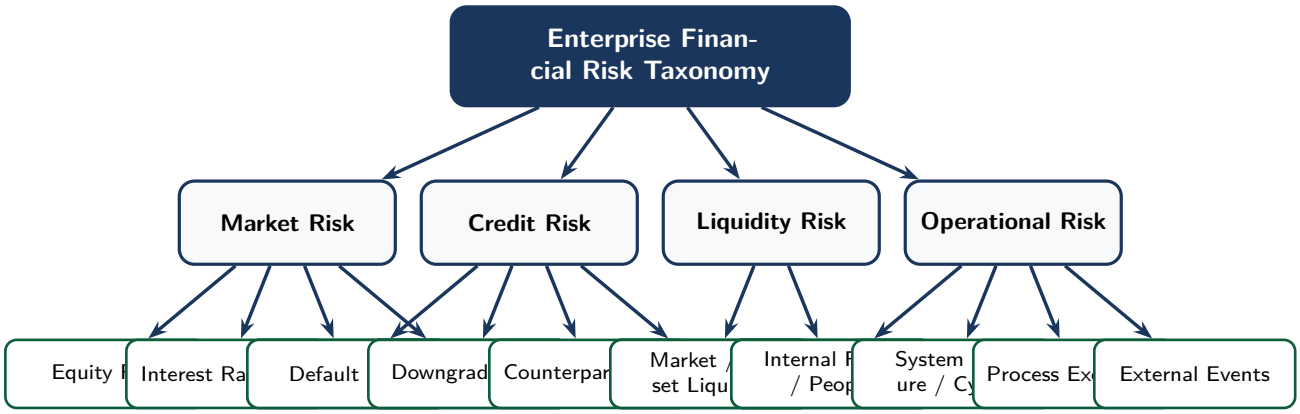


Figure 1.2: Comprehensive Enterprise Financial Risk Taxonomy Tree.

1. **Market Risk:** The risk of portfolio losses arising from adverse movements in observable financial market prices and parameters (equity prices, benchmark yield curves, foreign exchange rates, commodity prices, and implied volatilities).
2. **Credit Risk:** The economic loss suffered when an obligor, borrower, or swap counterparty fails to fulfill contractual debt obligations on time, or undergoes credit rating deterioration.
3. **Liquidity Risk:** Comprises two distinct dimensions:
   - *Market (Asset) Liquidity Risk:* The inability to liquidate large asset positions rapidly without incurring substantial price concessions or bid-ask slippage.
   - *Funding (Cash-Flow) Liquidity Risk:* The inability of an institution to meet immediate cash payment obligations, margin calls, or debt roll-overs, leading to insolvency.
4. **Operational Risk:** As defined by the Basel Committee on Banking Supervision (BCBS), the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events (including legal risk, cyber threats, and compliance sanctions, but excluding strategic and reputational risk).
5. **Sovereign and Systemic Risk:** The risk that a national government defaults on sovereign debt, institutes foreign exchange controls, or triggers a systemic contagion cascade that threatens the stability of the entire financial system.

1.3 Enterprise Risk Management and Governance Architecture

Traditional risk management operated in organizational "silos", where market risk, credit risk, insurance, and operational hazards were analyzed in isolation by disconnected departments. Modern financial institutions have transitioned to **Enterprise Risk Management (ERM)**, an integrated, holistic framework designed to evaluate firm-wide risk aggregation, capital allocation, and strategic risk-reward optimization.

1.3.1 The Three Lines of Defence Framework

Regulatory authorities worldwide (including the Reserve Bank of India, Federal Reserve, and European Central Bank) mandate the **Three Lines of Defence** governance structure to prevent conflicts of interest and ensure independent oversight.

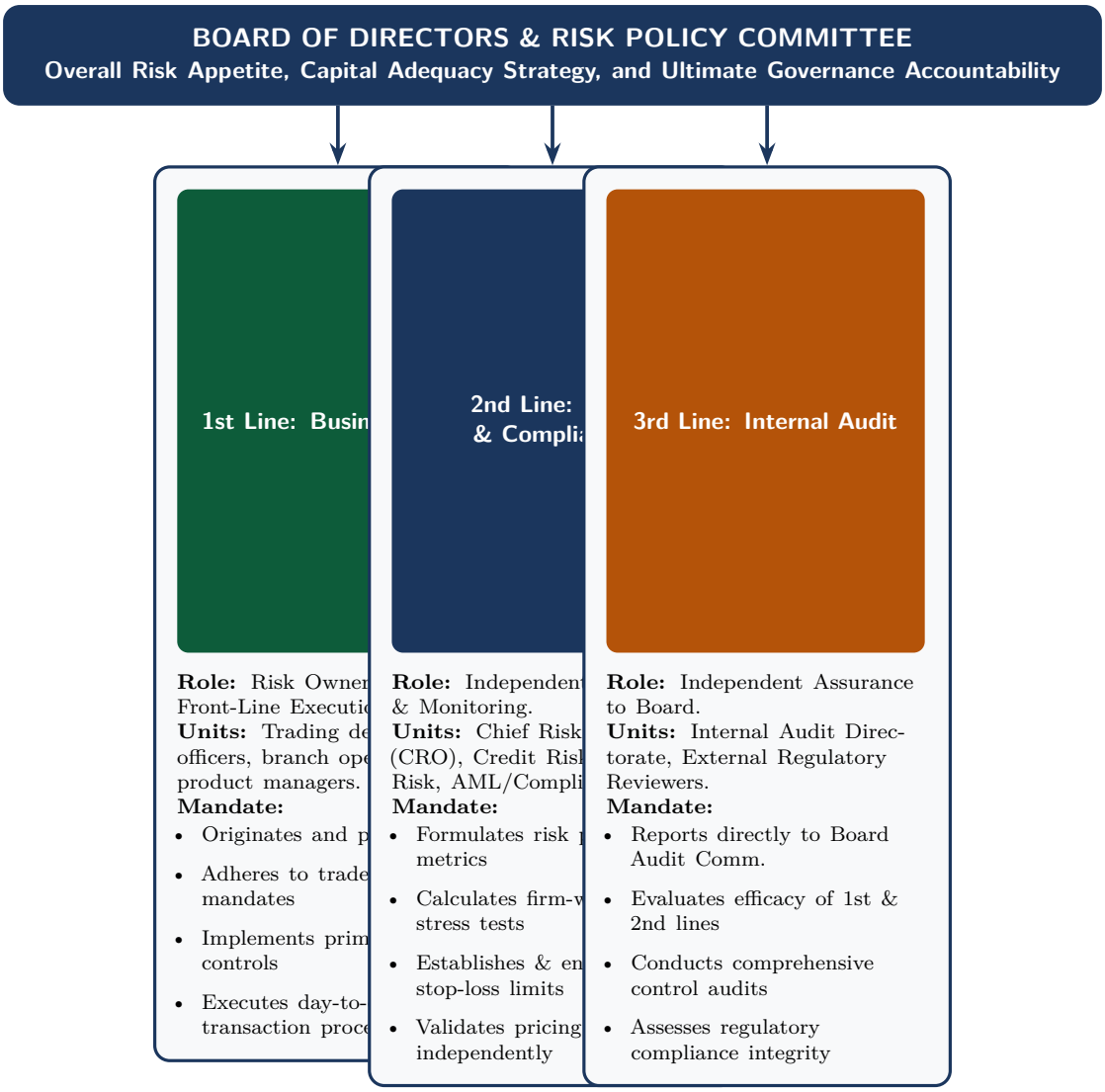


Figure 1.3: The Three Lines of Defence Governance Architecture (Rule 4 Symmetrical Connectors).

1.3.2 The Enterprise Risk Management (ERM) Process Cycle

In accordance with international risk management standards (ISO 31000 and the COSO ERM Framework), effective enterprise risk management follows a continuous, closed-loop feedback architecture:

1. **Context Establishment & Risk Appetite Definition:** Determining the organization's overarching risk tolerance, return objectives, and regulatory capital boundaries.
2. **Risk Identification:** Uncovering emerging exposures across all business lines through risk registers, process mapping, and Delphi expert panels.
3. **Risk Measurement & Assessment:** Quantifying probability distributions, Value at Risk, scenario analyses, and severity matrices.
4. **Risk Treatment & Mitigation:** Selecting suitable strategies: risk retention (capital reserves), risk transfer (insurance/derivatives), risk avoidance (exiting volatile business segments), or risk reduction (enhanced internal controls).

5. **Risk Monitoring & Escalation:** Continuous real-time limit monitoring, exception reporting to executive committees, and regulatory disclosures.

1.4 Asset-Class Specific Risk Profiles

Different financial asset classes exhibit fundamentally distinct cash flow dynamics, payoff functions, and risk sensitivities. Investors and portfolio managers must understand these characteristics to construct balanced portfolios and robust hedging strategies.

1.4.1 Risks of Commodity Investment

Unlike financial assets, commodities are consumption goods with physical delivery, storage, and decay constraints:

- **Cost of Carry Model:** Under continuous compounding, the theoretical forward price F_0 for a commodity with spot price S_0, risk-free rate r, storage cost percentage u, and convenience yield y is:
$$F_0 = S_0 e^{(r+u-y)T} \tag{1.5}$$
- **Convenience Yield (y):** The implicit benefit or embedded economic value of physically possessing an inventory of the commodity during an unexpected supply crunch.
- **Contango:** A market condition where futures prices trade higher than the spot price ($F_0 > S_0$), occurring when carrying costs exceed convenience yield ($r + u > y$). An investor rolling long futures contracts suffers a **negative roll yield** as the futures price converges downward to the spot price at expiration.
- **Backwardation (Normal Backwardation):** A market condition where futures prices trade at a discount to the spot price ($F_0 < S_0$), occurring when convenience yield is exceptionally high ($y > r + u$). Long futures holders earn a **positive roll yield** as futures prices converge upward toward spot.
- **Basis Risk:** The risk that the basis $b_t = S_t - F_t$ fluctuates unpredictably between hedge initiation and contract unwinding.

Example 1.4.1: Basis Risk and Roll Yield in Crude Oil Futures

An Indian airline requires 100,000 barrels of aviation turbine fuel (ATF) in three months. Because no liquid ATF contract exists on the Multi Commodity Exchange (MCX), the treasury hedges using MCX Crude Oil futures contracts (100 barrels per contract; 1,000 contracts total).

Market Data at $t = 0$:

- Current Spot Crude (S_0): Rs. 6,200 per barrel
- 3-Month Futures Contract ($F_{0,T}$): Rs. 6,450 per barrel
- Initial Basis: $b_0 = S_0 - F_0 = 6,200 - 6,450 = -$Rs. 250 (Market in Contango)

Market Data at $t = T$ (3 Months later):

- Spot Crude (S_T): Rs. 7,100 per barrel
- Actual ATF Spot Price Paid by Airline: Rs. 7,400 per barrel
- Futures Price at Maturity ($F_{T,T} = S_T$): Rs. 7,100 per barrel

Solution & Risk Breakdown:

1. Commodity Investment Risk

Core Drivers: Physical supply-demand disruptions, weather, geopolitical conflict, storage capacity.
Key Structural Traits:

- Carrying costs (u), storage, spoilage, insurance
- Convenience yield (y) from physical ownership
- Term structure: Contango vs Backwardation
- Rolling basis risk ($b_t = S_t - F_t$) on futures

2. Equity Investment Risk

Core Drivers: Corporate profitability, macroeconomic growth, competitive disruption, equity valuation multiples.
Key Structural Traits:

- Systematic market sensitivity (β) vs idiosyncratic shock
- Subordinated residual claim in corporate bankruptcy
- Dividend uncertainty & management agency cost
- Asymmetric volatility (leverage effect)

3. Fixed Income Investment Risk

Core Drivers: Central bank policy rates, inflation trajectory, credit quality migration, yield curve slope.
Key Structural Traits:

- Interest rate sensitivity: Duration (D^*) & Convexity (C)
- Credit spread widening & issuer default risk
- Reinvestment risk on periodic coupon payments
- Call / prepayment risk in declining rate environments

4. Currency (FX) Investment Risk

Core Drivers: Interest rate differentials, terms of trade, current account deficits, central bank FX intervention.
Key Structural Traits:

- Transaction exposure on contractual cash flows
- Translation exposure on overseas consolidated balances
- Operating (economic) exposure on competitive pricing
- Sovereign capital controls & convertibility barriers

Figure 1.4: Four-Pillar Asset Class Risk Comparison Grid (Rule 4 Symmetrical Dimensions).

1. **Futures Gain per barrel:**

$$\Delta F = F_T - F_0 = 7,100 - 6,450 = +\text{Rs. } 650 \text{ per barrel}$$

2. **Spot Price Increase for Crude:**

$$\Delta S = S_T - S_0 = 7,100 - 6,200 = +\text{Rs. } 900 \text{ per barrel}$$

3. **Roll Cost (Contango Drag):** The futures appreciated by Rs. 650, while spot rose by Rs. 900. The Rs. 250 drag reflects the initial contango penalty:

$$\text{Drag} = \Delta S - \Delta F = 900 - 650 = \text{Rs. } 250$$

4. **Cross-Commodity Basis Slippage:** The actual price paid for ATF rose from an initial equivalent of Rs. 6,400 to Rs. 7,400 (an increase of Rs. 1,000), leaving the airline exposed to an unhedged cross-commodity basis difference of Rs. $1,000 - \text{Rs. } 650 = \text{Rs. } 350$ per barrel.

1.4.2 Risks of Equity Investment

Equity investments represent residual ownership in a firm. Key risk dimensions include:

- **Market Beta Risk (β_i):** The sensitivity of the stock's return relative to changes in the broader benchmark index:

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} = \rho_{im}\frac{\sigma_i}{\sigma_m} \tag{1.6}$$

- **Asymmetric Leverage Effect:** When equity prices decline, financial leverage (Debt/Equity ratio) mechanically increases, making the company riskier and amplifying subsequent volatility.
- **Valuation and Multiple Compression Risk:** The vulnerability of stock prices to contractions in price-to-earnings (P/E) or enterprise value multiples caused by rising risk-free rates or macroeconomic slowdowns.

Example 1.4.2: Portfolio Beta and Factor Exposure on NSE Nifty 50

An institutional portfolio manager holds an equity portfolio valued at Rs. 50,00,00,000 distributed across three sectors:

Sector Allocation	Portfolio Weight (w_i)	Value (Rs. Crores)	Sector Beta (β_i)
Banking & Financial Services	40%	Rs. 20.0 Cr	1.35
Information Technology (IT)	35%	Rs. 17.5 Cr	0.90
Fast Moving Consumer Goods (FMCG)	25%	Rs. 12.5 Cr	0.60

Task:

1. Calculate the overall portfolio beta (β_P).
2. Estimate the expected portfolio value drop if the benchmark Nifty 50 Index falls by 4.5% due to unexpected geopolitical tensions.

Solution:

1. **Portfolio Beta Calculation:**

$$\beta_P = \sum_{i=1}^{3} w_i \beta_i = (0.40 \times 1.35) + (0.35 \times 0.90) + (0.25 \times 0.60) = 0.540 + 0.315 + 0.150 = 1.005$$

2. **Expected Portfolio Percentage Decline:**

$$\Delta R_P \approx \beta_P \times \Delta R_m = 1.005 \times (-4.5\%) = -4.5225\%$$

3. **Expected Value Loss:**

$$\text{Loss} = \text{Rs. } 50,00,00,000 \times 0.045225 = \text{Rs. } 2,26,12,500 \quad (\approx \text{Rs. } 2.261 \text{ Crores})$$

1.4.3 Risks of Fixed Income Investment

Fixed income risk is driven by changes in the yield curve and issuer creditworthiness:

- **Macaulay Duration (D):** The weighted average maturity of the bond's cash flows, where the weights are the present values of each cash flow:

$$D = \frac{1}{P} \sum_{t=1}^{n} \frac{t \cdot C_t}{(1+y)^t} \tag{1.7}$$

- **Modified Duration (D^*):** Measures the percentage price sensitivity with respect to a change in yield:

$$D^* = \frac{D}{1+y} = -\frac{1}{P} \frac{\mathrm{d}P}{\mathrm{d}y} \tag{1.8}$$

- **Convexity (C):** The second derivative of price with respect to yield, capturing curvature:

$$C = \frac{1}{P} \frac{\mathrm{d}^2 P}{\mathrm{d}y^2} = \frac{1}{P(1+y)^2} \sum_{t=1}^{n} \frac{(t^2+t)C_t}{(1+y)^t} \tag{1.9}$$

- **Second-Order Price Approximation:** Combining modified duration and convexity:

$$\frac{\Delta P}{P} \approx -D^* \Delta y + \frac{1}{2} C (\Delta y)^2 \tag{1.10}$$

- **DV01 (Dollar Value of an 01 / PVBP):** The absolute monetary loss resulting from a 1 basis point ($0.01\% = 0.0001$) increase in yield:

$$\text{DV01} = D^* \times P \times 0.0001 \tag{1.11}$$

Example 1.4.3: Bond Duration, Convexity, and RBI Repo Rate Shock

A commercial bank's treasury holds an Indian Government Bond (G-Sec) portfolio with a total market value of Rs. 100 Crores (Rs. 1,00,00,00,000). The portfolio has a weighted modified duration of $D^* = 6.40$ years and a convexity of $C = 58.0$ years^2.
The Reserve Bank of India (RBI) unexpectedly hikes the policy repo rate by 75 bps ($\Delta y = +0.0075$).
Task:

1. Compute the portfolio DV01.
2. Estimate the percentage and total rupee value change using first-order duration alone.
3. Refine the calculation incorporating the second-order convexity correction.

Solution:

1. **DV01 Calculation:**

$$\text{DV01} = D^* \times P \times 0.0001 = 6.40 \times \text{Rs. } 1,00,00,00,000 \times 0.0001 = \text{Rs. } 6,40,000 \text{ per basis point}$$

2. **First-Order Duration Approximation ($\Delta y = +75$ bps):**

$$\frac{\Delta P}{P} \approx -D^* \Delta y = -6.40 \times 0.0075 = -0.0480 \quad (-4.80\%)$$

$$\Delta P = \text{Rs. } 100 \text{ Cr} \times (-0.0480) = -\text{Rs. } 4.80 \text{ Crores}$$

3. **Second-Order Convexity Refinement:**

$$\text{Convexity Adjustment} = \frac{1}{2} C (\Delta y)^2 = \frac{1}{2} \times 58.0 \times (0.0075)^2 = 0.5 \times 58.0 \times 0.00005625 = +0.001631 \quad (+0.163$$

$$\frac{\Delta P_{\text{total}}}{P} \approx -4.80\% + 0.1631\% = -4.6369\%$$

$$\Delta P_{\text{refined}} = \text{Rs. } 100 \text{ Cr} \times (-0.046369) = -\text{Rs. } 4.6369 \text{ Crores}$$

The positive convexity softens the portfolio loss by $\approx$ Rs. 16.31 Lakhs.

1.4.4 Risks of Currency Investment

Multinational corporations and foreign investors face foreign exchange (FX) volatility driven by interest rate differentials, inflation, and cross-border trade balances:

- **Transaction Risk:** The risk of exchange rate fluctuations between entering into a cross-border contract and settling the cash flows.
- **Translation Risk:** Accounting gains or losses when consolidating financial statements of foreign subsidiaries into the domestic reporting currency.
- **Economic (Operating) Risk:** The structural impact of permanent currency depreciations or appreciations on a firm's long-term export competitiveness, supply chain costs, and future cash flows.
- **Covered Interest Parity (CIP):** Prevents arbitrage between spot and forward currency markets:

$$F = S \times \frac{1 + r_d}{1 + r_f} \tag{1.12}$$

where S is the spot rate (Domestic/Foreign), F is the forward rate, r_d is the domestic risk-free rate, and r_f is the foreign risk-free rate.

Example 1.4.4: Covered Interest Parity and FX Forward Hedging

An Indian IT services firm expects to receive an export receivable of USD $10,000,000$ in 6 months ($T = 0.5$ years).

- Current Spot Rate (S_0): Rs. 83.50 per USD
- Indian 6-month risk-free rate (r_{INR}): 6.50% p.a.
- US 6-month risk-free rate (r_{USD}): 5.00% p.a.

Task:

1. Compute the theoretical 6-month forward rate $F_{0,0.5}$ under Covered Interest Parity.
2. Determine the locked-in rupee cash flow if the firm executes a forward contract to eliminate currency risk.

Solution:

1. **Theoretical Forward Rate (CIP):**

$$F = S_0 \times \left[\frac{1 + r_{\text{INR}} \times 0.5}{1 + r_{\text{USD}} \times 0.5}\right] = 83.50 \times \left[\frac{1 + (0.065 \times 0.5)}{1 + (0.050 \times 0.5)}\right] = 83.50 \times \left[\frac{1.0325}{1.0250}\right]$$

$$F = 83.50 \times 1.007317 = \text{Rs. } 84.1109 \text{ per USD}$$

2. **Forward Hedge Rupee Realization:**

   Rupee Cash Flow = USD $10,000,000 \times$ Rs. 84.1109 = Rs. $84,11,09,000$ (Rs. 84.11 Crores)

   By locking in the forward contract, the exporter fully eliminates the downside risk of USD depreciation against the Indian Rupee over the 6-month interval.

1.5 Market Risk Measurement Framework

Systematic market risk measurement requires aggregate risk metrics that translate multidimensional market exposures into a single, actionable capital reserve figure. The evolution of market risk measurement spans three eras:

1. **Sensitivity Metrics (The Greeks, Beta, Duration):** Measures first- and second-order price changes with respect to isolated risk factors ($\Delta, \Gamma, \mathcal{V}, D^*, \beta$). While computationally transparent, they cannot be directly aggregated across asset classes (e.g., adding bond duration to option gamma).
2. **Value at Risk (VaR):** Introduced in the mid-1990s (J.P. Morgan RiskMetrics), VaR unified portfolio risk into a single statistical currency loss metric at a specified confidence level over a defined holding period.
3. **Coherent Risk Measures & Expected Shortfall (ES):** Developed to overcome the mathematical deficiencies of VaR (specifically its failure to satisfy subadditivity and its tail blindness beyond the confidence percentile), leading to the Basel Committee's Fundamental Review of the Trading Book (FRTB) reforms.

Chapter 2

Value at Risk (VaR) Methods and Hedging

Lecture
21/08/2026

While sensitivity metrics (such as duration, convexity, and the option Greeks) capture localized price responses to isolated market factors, institutional portfolio managers require a unified statistical framework to quantify aggregate downside exposure. This chapter examines the theoretical foundations, computational methodologies, and practical applications of **Value at Risk (VaR)** and **Expected Shortfall (ES)**, followed by linear and non-linear risk hedging strategies.

2.1 Overview of Value at Risk (VaR)

Definition 2.1.1: Value at Risk (VaR)

For a given portfolio, time horizon T, and confidence level $\alpha \in (0,1)$, the **Value at Risk** (VaR_α) represents the maximum expected loss that will not be exceeded with probability α over the horizon T.
Formally, letting $L = -\Delta\Pi$ denote the loss variable (where $\Delta\Pi = \Pi_T - \Pi_0$ is the change in portfolio value), VaR_α is the $(1-\alpha)$ upper quantile of the loss distribution:

$$\mathbb{P}(L > \text{VaR}_\alpha) = 1 - \alpha \quad \iff \quad \mathbb{P}(\Delta\Pi \leq -\text{VaR}_\alpha) = 1 - \alpha \tag{2.1}$$

Equivalently, in terms of the cumulative distribution function (CDF) of portfolio returns $F_R(r)$:

$$\text{VaR}_\alpha = -V_0 \cdot F_R^{-1}(1-\alpha) \tag{2.2}$$

where V_0 is the current portfolio market value.

In banking and trading desk operations, typical parameters are:

- **Internal Risk Management:** $\alpha = 95\%$ or 99% over a 1-day holding period ($T = 1$).
- **Basel Regulatory Market Risk Capital:** $\alpha = 99\%$ over a 10-day holding period ($T = 10$).
- **Credit / Economic Capital:** $\alpha = 99.9\%$ over a 1-year horizon ($T = 252$).

2.1.1 Time Horizon Scaling: The Square-Root-of-Time Rule

Under the assumption that daily portfolio returns are independent and identically distributed (i.i.d.) normal random variables with constant mean and volatility, the T-day volatility scales as

$\sigma_T = \sigma_1\sqrt{T}$. Consequently, the T-day VaR scales according to:

$$\text{VaR}_{\alpha,T} = \text{VaR}_{\alpha,1} \times \sqrt{T} \tag{2.3}$$

Limitations of the $\sqrt{T}$ Scaling Rule

The square-root-of-time rule strictly requires: (i) zero serial autocorrelation in returns, and (ii) stability of the underlying distribution. When returns exhibit volatility clustering (GARCH effects) or mean reversion, the $\sqrt{T}$ rule either understates or overstates long-horizon risk.

2.1.2 Deficiencies of VaR and Coherent Risk Measures

Despite its widespread regulatory adoption, VaR suffers from two mathematical and practical limitations:

1. **Non-Subadditivity:** A risk measure $\rho(\cdot)$ is subadditive if $\rho(A+B) \leq \rho(A)+\rho(B)$, reflecting the principle that diversification reduces risk. For non-normal return distributions (e.g., portfolios with short option positions or credit default risk), VaR can violate subadditivity:

$$\text{VaR}_\alpha(A+B) > \text{VaR}_\alpha(A) + \text{VaR}_\alpha(B) \tag{2.4}$$

   Consequently, VaR is **not** a coherent risk measure in the axiomatic sense of Artzner, Delbaen, Eber, and Heath (1999).

2. **Tail Blindness:** VaR only defines the boundary quantile threshold. It reveals nothing about the magnitude or severity of losses in the catastrophic $(1-\alpha)$ tail beyond the cutoff.

2.1.3 Expected Shortfall (ES / Conditional VaR)

To remedy these deficiencies, the Basel III framework (under the Fundamental Review of the Trading Book - FRTB) replaced the 99% 10-day VaR with the 97.5% Expected Shortfall.

Definition 2.1.2: Expected Shortfall (Conditional VaR)

The **Expected Shortfall** (ES_α) at confidence level α is the conditional expectation of loss given that the loss exceeds VaR_α:

$$\text{ES}_\alpha = \mathbb{E}\left[L \mid L \geq \text{VaR}_\alpha\right] = \frac{1}{1-\alpha}\int_\alpha^1 \text{VaR}_u \, du \tag{2.5}$$

For a continuous distribution with loss probability density function $f_L(x)$:

$$\text{ES}_\alpha = \frac{1}{1-\alpha}\int_{\text{VaR}_\alpha}^{\infty} x \cdot f_L(x)\, dx \tag{2.6}$$

ES_α is subadditive, convex, and coherent under all underlying return distributions.

Example 2.1.1: VaR Horizon Scaling and Confidence Level Conversion

A proprietary equity trading desk at an Indian investment bank manages a Rs. 20,00,00,000 (Rs. 20 Crores) portfolio tracking the BSE Sensex. The estimated daily return standard deviation is $\sigma_{\text{daily}} = 1.25\%$. Assume daily returns are normally distributed with zero mean

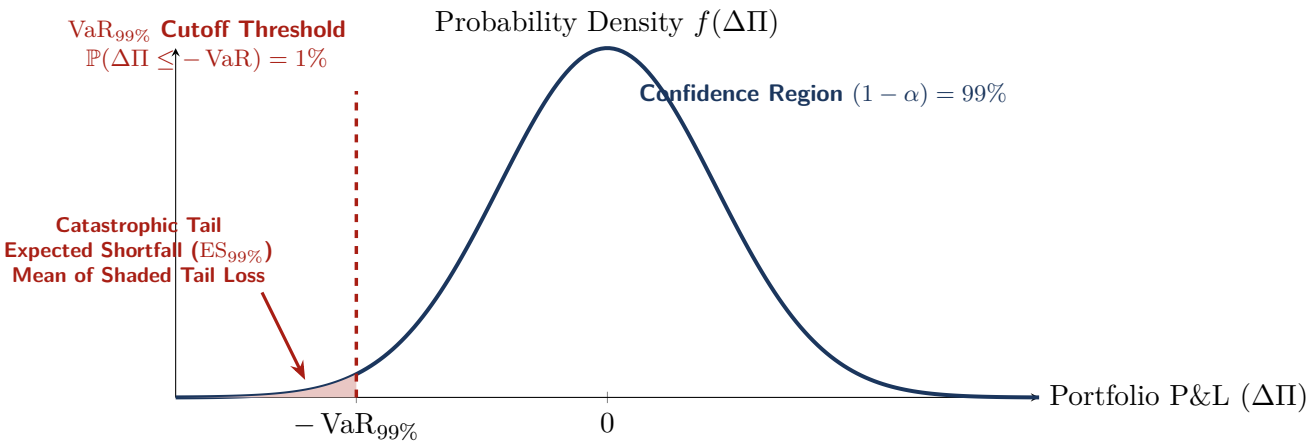


Figure 2.1: Portfolio Return Distribution: VaR_α Threshold and Shaded Expected Shortfall Tail.

($\mu = 0$).
Task:

1. Compute the 1-day 95% VaR ($z_{0.95} = 1.645$) and 1-day 99% VaR ($z_{0.99} = 2.326$).
2. Compute the 10-day 99% regulatory capital VaR using the square-root-of-time rule.
3. For a standard normal distribution, $\text{ES}_{99\%} = \frac{\phi(z_{0.99})}{1-0.99}\sigma = \frac{\phi(2.326)}{0.01}\sigma \approx 2.665 \times \sigma$. Calculate the 1-day 99% Expected Shortfall.

Solution:

1. **1-Day VaR Calculation:**

   $$\text{VaR}_{95\%,\text{1-day}} = V_0 \times z_{0.95} \times \sigma_{\text{daily}} = \text{Rs. } 20,00,00,000 \times 1.645 \times 0.0125 = \text{Rs. } 41,12,500 \quad (\approx \text{Rs. } 41.13 \text{ Lakhs})$$

   $$\text{VaR}_{99\%,\text{1-day}} = V_0 \times z_{0.99} \times \sigma_{\text{daily}} = \text{Rs. } 20,00,00,000 \times 2.326 \times 0.0125 = \text{Rs. } 58,15,000 \quad (\approx \text{Rs. } 58.15 \text{ Lakhs})$$

2. **10-Day 99% Regulatory VaR:**

   $$\text{VaR}_{99\%,\text{10-day}} = \text{VaR}_{99\%,\text{1-day}} \times \sqrt{10} = \text{Rs. } 58,15,000 \times 3.162278 = \text{Rs. } 1,83,88,645 \quad (\approx \text{Rs. } 1.84 \text{ Crores})$$

3. **1-Day 99% Expected Shortfall:**

   $$\text{ES}_{99\%,\text{1-day}} = V_0 \times 2.665 \times \sigma_{\text{daily}} = \text{Rs. } 20,00,00,000 \times 2.665 \times 0.0125 = \text{Rs. } 66,62,500 \quad (\approx \text{Rs. } 66.63 \text{ Lakhs})$$

   When losses breach the 99% threshold of Rs. 58.15 Lakhs, the conditional average loss is Rs. 66.63 Lakhs.

2.2 VaR Estimation Methodologies

VaR estimation approaches are fundamentally bifurcated into **Local Valuation** and **Full Valuation** paradigms.

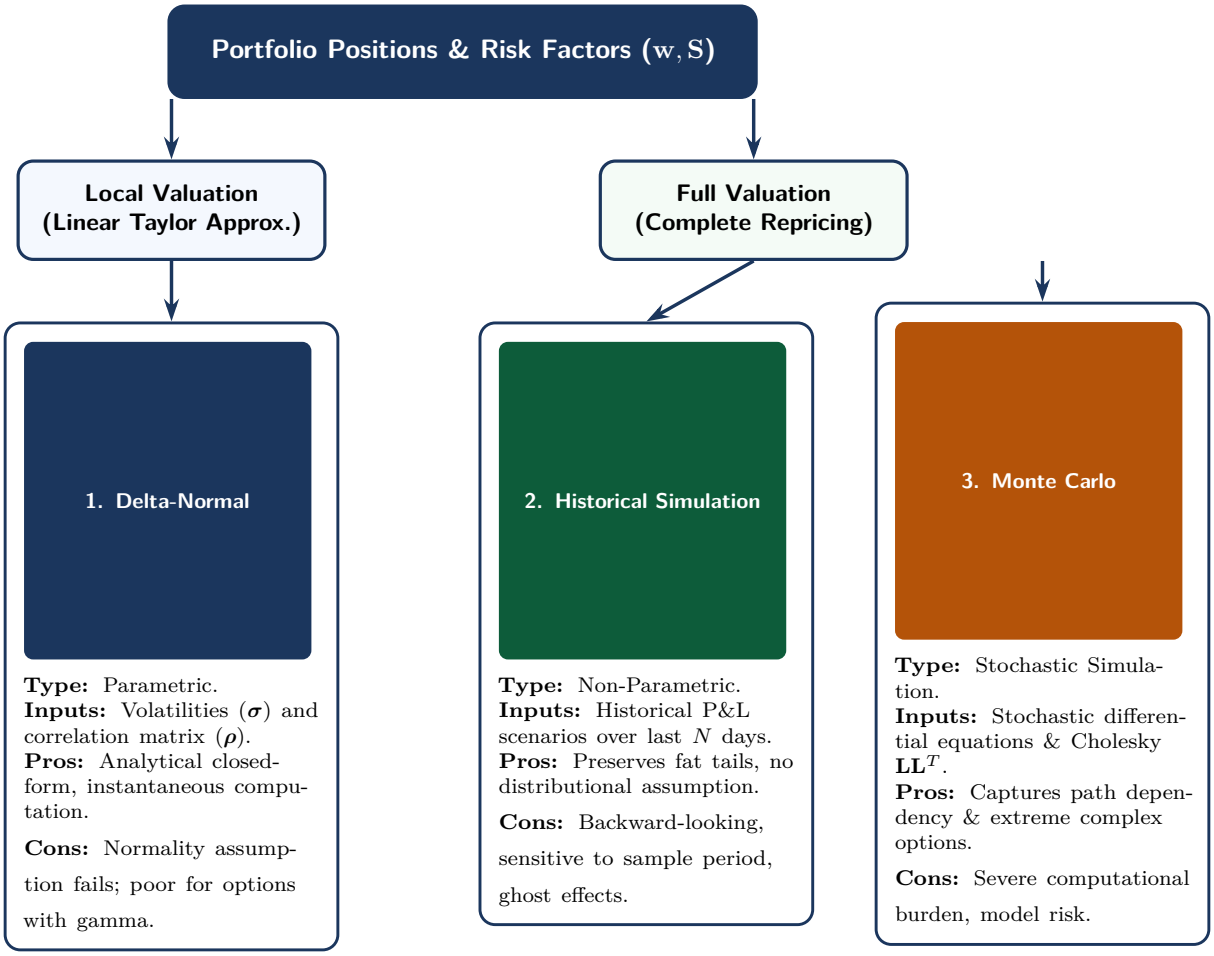


Figure 2.2: VaR Estimation Methodologies Architecture (Rule 4 Clearance and Symmetry Standards).

2.2.1 1. Delta-Normal (Parametric / Variance-Covariance) Method

The Delta-Normal approach assumes that the underlying risk factors follow a multivariate normal distribution. For an n-asset linear portfolio with value V_0, weight vector $\mathbf{w} = (w_1, w_2, \ldots, w_n)^T$, and variance-covariance matrix $\mathbf{\Sigma}$:

$$\sigma_P^2 = \mathbf{w}^T \mathbf{\Sigma} \mathbf{w} = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij} = \sum_{i=1}^{n} w_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n} \sum_{j<i} w_i w_j \rho_{ij} \sigma_i \sigma_j \tag{2.7}$$

The parametric VaR is:

$$\text{VaR}_\alpha = z_\alpha \cdot \sigma_P \cdot V_0 \tag{2.8}$$

where z_α is the standard normal critical value ($z_{0.95} = 1.645$, $z_{0.99} = 2.326$).

Example 2.2.1: Two-Asset Portfolio Parametric VaR and Diversification Benefit

An asset management firm holds a two-asset portfolio valued at Rs. 10,00,00,000 (Rs. 10 Crores):

- Asset 1: Large-cap Equity ETF ($w_1 = 0.60$, $V_1 = $ Rs. 6 Cr), Daily $\sigma_1 = 1.40\%$
- Asset 2: Long-Term G-Sec Bond ETF ($w_2 = 0.40$, $V_2 = $ Rs. 4 Cr), Daily $\sigma_2 = 0.60\%$
- Correlation Coefficient: $\rho_{12} = -0.20$

Task:

1. Calculate the individual 1-day 99% VaR of Asset 1 and Asset 2 independently.
2. Calculate the diversified portfolio daily volatility σ_P and diversified 1-day 99% VaR ($z_{0.99} = 2.326$).
3. Quantify the diversification benefit (Rs. savings).

Solution:

1. **Individual VaR Figures:**

$$\text{VaR}_1 = 2.326 \times 0.0140 \times \text{Rs. } 6,00,00,000 = \text{Rs. } 19,53,840 \quad (\approx \text{Rs. } 19.54 \text{ Lakhs})$$

$$\text{VaR}_2 = 2.326 \times 0.0060 \times \text{Rs. } 4,00,00,000 = \text{Rs. } 5,58,240 \quad (\approx \text{Rs. } 5.58 \text{ Lakhs})$$

Undiversified Total VaR $= \text{VaR}_1 + \text{VaR}_2 = 19,53,840 + 5,58,240 = \text{Rs. } 25,12,080$ (Rs. 25.12 Lakhs)

2. **Portfolio Volatility σ_P:**

$$\begin{aligned}
\sigma_P^2 &= w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2 \\
&= (0.6)^2(0.014)^2 + (0.4)^2(0.006)^2 + 2(0.6)(0.4)(-0.20)(0.014)(0.006) \\
&= (0.36)(0.000196) + (0.16)(0.000036) - 0.096(0.000084) \\
&= 0.00007056 + 0.00000576 - 0.000008064 = 0.000068256 \\
\sigma_P &= \sqrt{0.000068256} = 0.0082617 \quad (0.8262\%)
\end{aligned}$$

$\text{VaR}_{\text{diversified}} = 2.326 \times \sigma_P \times V_0 = 2.326 \times 0.0082617 \times \text{Rs. } 10,00,00,000 = \text{Rs. } 19,21,672$ ($\approx$ Rs. 19.22 Lakh

3. **Diversification Benefit:**

Benefit = Undiversified VaR − Diversified VaR $= \text{Rs. } 25,12,080 - \text{Rs. } 19,21,672 = \text{Rs. } 5,90,408$ ($\approx$ Rs. 5.

Negative correlation ($\rho = -0.20$) generates a 23.5% risk reduction.

2.2.2 2. Historical Simulation Method

Historical Simulation is a non-parametric full-valuation technique. Given M historical trading days:

1. Identify current market risk factor prices $\mathbf{S}_0 = (S_{1,0}, S_{2,0}, \ldots, S_{k,0})$.
2. Collect past daily returns $\mathbf{r}_t$ for $t = 1, 2, \ldots, M$.
3. Generate M hypothetical scenario prices: $S^*_{i,t} = S_{i,0} \times (1 + r_{i,t})$.
4. Fully revalue the portfolio under each scenario: $\Pi^*_t = V(\mathbf{S}^*_t)$.
5. Compute simulated P&L values: $\Delta\Pi_t = \Pi^*_t - \Pi_0$.
6. Sort the P&L array from most severe loss to largest gain: $\Delta\Pi_{(1)} \leq \Delta\Pi_{(2)} \leq \cdots \leq \Delta\Pi_{(M)}$.
7. The α-confidence VaR is the $(1-\alpha) \times M$-th worst loss.

Example 2.2.2: Historical Simulation Quantile Extraction

A risk manager uses $M = 500$ daily historical return scenarios to compute the 1-day 99% VaR for an option portfolio.
Task:

1. Identify which ranked observation index corresponds to the 99% VaR.
2. Given the 8 worst simulated P&L results (Rs.):

Rank (k)	1	2	3	4	5	6	7	8
P&L (Rs.)	−18.5 L	−15.2 L	−13.8 L	−12.4 L	−11.5 L	−10.9 L	−9.8 L	−9.2 L

Determine the 99% 1-day VaR and the empirical Expected Shortfall $\text{ES}_{99\%}$.

Solution:

1. **Quantile Observation Index:**

$$k = (1-\alpha) \times M = (1 - 0.99) \times 500 = 5$$

The 5th worst loss defines the 99% VaR threshold.

2. **Empirical VaR:**

$$\text{VaR}_{99\%} = -\Delta\Pi_{(5)} = \text{Rs. } 11,50,000 \quad (\text{Rs. } 11.50 \text{ Lakhs})$$

3. **Empirical Expected Shortfall ($\text{ES}_{99\%}$):** The arithmetic mean of all losses equal to or exceeding the 5th worst loss (Ranks 1 to 5):

$$\text{ES}_{99\%} = \frac{1}{5}\sum_{k=1}^{5}(-\Delta\Pi_{(k)}) = \frac{18.5 + 15.2 + 13.8 + 12.4 + 11.5}{5} = \frac{71.4}{5} = \text{Rs. } 14,28,000 \quad (\text{Rs. } 14.28 \text{ Lakhs})$$

2.2.3 3. Monte Carlo Simulation Method

Monte Carlo simulation models future market evolutions by generating thousands of pseudo-random sample paths based on continuous-time stochastic processes. For k correlated risk factors

driven by Geometric Brownian Motion:

$$\mathrm{d}S_{i,t} = \mu_i S_{i,t}\,\mathrm{d}t + \sigma_i S_{i,t}\,\mathrm{d}W_{i,t} \tag{2.9}$$

where $\mathbb{E}[\mathrm{d}W_{i,t}\mathrm{d}W_{j,t}] = \rho_{ij}\mathrm{d}t$.

To simulate correlated Wiener increments, the covariance matrix $\mathbf{\Sigma}$ is factored via **Cholesky decomposition**:

$$\mathbf{\Sigma} = \mathbf{L}\mathbf{L}^T \tag{2.10}$$

where $\mathbf{L}$ is a lower triangular matrix. Given independent standard normal variates $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, correlated shocks are generated as:

$$\boldsymbol{\epsilon} = \mathbf{L}\mathbf{Z} \tag{2.11}$$

Each path produces terminal prices $\mathbf{S}_T^{(j)}$, the portfolio is fully repriced, and VaR is extracted from the simulated empirical distribution.

2.2.4 Methodological Comparison Matrix

Evaluation Dimension	Delta-Normal (Parametric)	Historical Simulation	Monte Carlo Simulation
Valuation Paradigm	Local Valuation (Linear)	Full Valuation (Exact)	Full Valuation (Exact)
Distributional Bias	Assumes Multivariate Normal	Empirical (Distribution-Free)	Explicit Stochastic Model
Non-Linear Options	Inaccurate (Ignores Gamma)	Accurate	Highly Accurate
Fat-Tail Capture	Poor	Excellent	Depends on Specified SDE
Computational Cost	Negligible (Analytical closed)	Moderate	Very High (CPU/GPU Heavy)
Primary Vulnerability	Model risk (Normality failure)	Ghost features & past bias	Model calibration risk

Table 2.1: Comparative Synthesis of Value at Risk Estimation Methodologies.

2.3 Stress Testing and Scenario Analysis

Because VaR calibrates to "normal" market fluctuations, regulatory standards require institutions to execute comprehensive stress testing programs to evaluate solvency during extreme systemic shocks:

- **Historical Scenario Testing:** Simulates the exact risk factor price trajectories observed during historical crises:
  - *1987 Black Monday:* S&P 500 plunged 22.6% in a single day.
  - *1998 Russian Default / LTCM Collapse:* Sovereign default and extreme flight-to-liquidity.
  - *2008 Global Financial Crisis:* Lehman bankruptcy, interbank liquidity freeze, credit spread blowout.
  - *March 2020 COVID Shock:* Global equities collapsed 35%; oil traded briefly at negative prices.
- **Hypothetical Scenario Testing:** Simulates forward-looking structural breaks without historical precedent (e.g., severe emerging market sovereign debt default, major cyber disruption of global payments, regional military escalations).

- **Reverse Stress Testing:** Identifies specific combinations of adverse market shocks that would completely exhaust an institution's regulatory capital buffer, causing failure.

2.4 Hedging Strategies: Linear Risk

Hedging linear risk involves utilizing futures, forwards, or swaps whose payoff is a linear function of the underlying asset price.

2.4.1 Basis Risk in Futures Hedging

Let S_1, F_1 denote spot and futures prices at hedge inception (t_1), and S_2, F_2 denote prices at hedge termination (t_2). The **basis** at any time t is defined as:

$$b_t = S_t - F_t \tag{2.12}$$

For a short hedger (holding long spot asset and selling futures), the effective price realized per unit at t_2 is:

$$P_{\text{eff}} = S_2 + (F_1 - F_2) = F_1 + (S_2 - F_2) = F_1 + b_2 \tag{2.13}$$

If basis $b_2 = S_2 - F_2$ is known with certainty at inception (e.g., when closing at contract maturity where $b_T = 0$), the hedge is perfect. If b_2 fluctuates unpredictably, the hedger faces **basis risk**.

2.4.2 Optimal Hedging and Minimum-Variance Hedge Ratio

Let h denote the hedge ratio (units of futures per unit of spot asset). The change in the value of the hedged portfolio $\Delta\Pi$ over time interval Δt is:

$$\Delta\Pi = \Delta S - h\,\Delta F \tag{2.14}$$

The variance of the hedged position is:

$$\operatorname{Var}(\Delta\Pi) = \sigma_S^2 + h^2\sigma_F^2 - 2h\operatorname{Cov}(\Delta S, \Delta F) \tag{2.15}$$

To determine the **minimum-variance hedge ratio** h^*, take the first derivative with respect to h and set it to zero:

$$\frac{\mathrm{d}\operatorname{Var}(\Delta\Pi)}{\mathrm{d}h} = 2h\sigma_F^2 - 2\operatorname{Cov}(\Delta S, \Delta F) = 0 \tag{2.16}$$

Solving for h^* yields:

$$h^* = \frac{\operatorname{Cov}(\Delta S, \Delta F)}{\sigma_F^2} = \rho_{SF}\frac{\sigma_S}{\sigma_F} \tag{2.17}$$

Hedge Ratio as an OLS Regression Coefficient

The optimal hedge ratio h^* is mathematically identical to the Ordinary Least Squares (OLS) regression slope coefficient obtained by regressing historical changes in spot prices (ΔS) on changes in futures prices (ΔF):

$$\Delta S_t = \alpha + h^*\Delta F_t + \epsilon_t \tag{2.18}$$

The proportion of risk eliminated by the optimal hedge is measured by the coefficient of determination R^2:

$$\text{Hedging Effectiveness} = 1 - \frac{\operatorname{Var}(\Delta\Pi)}{\operatorname{Var}(\Delta S)} = \rho_{SF}^2 = R^2 \tag{2.19}$$

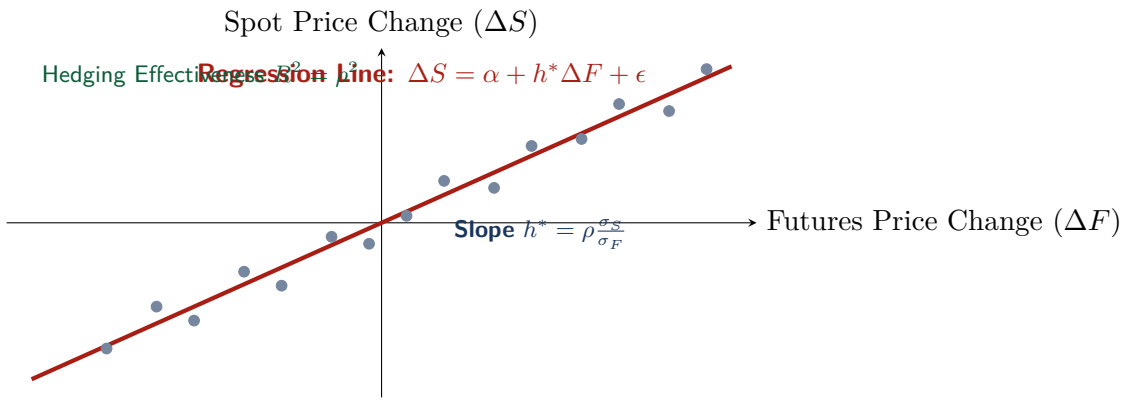


Figure 2.3: Ordinary Least Squares (OLS) Minimum-Variance Hedge Ratio Regression Diagram.

The optimal number of futures contracts N^* required to hedge an exposure of size Q_A using futures contracts with unit size Q_F is:

$$N^* = h^* \frac{Q_A}{Q_F} \tag{2.20}$$

For an equity portfolio of value V_P hedged using index futures with index level I and multiplier M (so futures contract value $V_F = I \times M$):

$$N^* = \beta_P \frac{V_P}{V_F} \tag{2.21}$$

Example 2.4.1: Optimal Hedge Ratio and Index Futures Contract Sizing

A mutual fund holds an Indian equity portfolio valued at Rs. 25,00,00,000 (Rs. 25 Crores) with a portfolio beta of $\beta_P = 1.20$. The manager expects a near-term market correction and decides to hedge the portfolio over the next 2 months using NSE Nifty 50 Index Futures contracts.

Market Parameters:

- Current Nifty 50 Futures Index Level (I): $24,500$
- Nifty 50 Futures Market Lot Size (M): 25 shares per contract
- Contract Notional Value: $V_F = 24,500 \times 25 = \text{Rs. } 6,12,500$

Task:

1. Calculate the number of Nifty futures contracts required to completely neutralize market risk ($\beta \to 0$).
2. Determine the number of contracts required if the manager wishes to reduce portfolio beta from 1.20 to a defensive target of $\beta^* = 0.40$.
3. If the Nifty Index drops by 5% while the hedge is active, verify the net portfolio performance.

Solution:

1. **Complete Hedge ($\beta \to 0$):**

$$N^* = \beta_P \frac{V_P}{V_F} = 1.20 \times \frac{\text{Rs. } 25,00,00,000}{\text{Rs. } 6,12,500} = 1.20 \times 408.163 = 489.79 \approx 490 \text{ Contracts (Short)}$$

2. **Target Beta Reduction ($\beta_P \to \beta^* = 0.40$):**

$$N^* = (\beta_P - \beta^*)\frac{V_P}{V_F} = (1.20 - 0.40) \times \frac{\text{Rs. } 25,00,00,000}{\text{Rs. } 6,12,500} = 0.80 \times 408.163 = 326.53 \approx 327 \text{ Contracts (Short)}$$

3. **Verification for Complete Hedge under a -5% Index Fall:**

- Unhedged Equity Drop: $\Delta V_P = \beta_P \times \Delta R_m \times V_P = 1.20 \times (-0.05) \times \text{Rs. } 25 \text{ Cr} = -\text{Rs. } 1.50 \text{ Crores}$
- Futures Gain (490 short contracts):

$$\text{Points Drop per Contract} = 24,500 \times 0.05 = 1,225 \text{ points}$$

$$\text{Gain per Contract} = 1,225 \times 25 = \text{Rs. } 30,625$$

$$\text{Total Futures Profit} = 490 \times \text{Rs. } 30,625 = +\text{Rs. } 1,50,06,250 \quad (\approx +\text{Rs. } 1.5006 \text{ Crores})$$

- Net Position: $-\text{Rs. } 1.5000 \text{ Cr} + \text{Rs. } 1.5006 \text{ Cr} \approx +\text{Rs. } 6,250$ (Market loss successfully neutralized).

2.5 Hedging Strategies: Non-Linear Risk

Options possess non-linear, asymmetric payoff profiles. Consequently, their price sensitivity varies continuously with underlying price movements, time decay, and implied volatility.

2.5.1 The Greeks and Second-Order Taylor Expansion

Let $V(S, t, \sigma)$ represent the theoretical price of an option. The fundamental Greeks are:

$$\Delta = \frac{\partial V}{\partial S}, \qquad \Gamma = \frac{\partial^2 V}{\partial S^2}, \qquad \mathcal{V} = \frac{\partial V}{\partial \sigma}, \qquad \Theta = \frac{\partial V}{\partial t} \tag{2.22}$$

The change in option value ΔV over small changes in underlying price ΔS, time Δt, and volatility $\Delta\sigma$ is approximated via a second-order Taylor series expansion:

$$\Delta V \approx \Delta \cdot \Delta S + \frac{1}{2}\Gamma(\Delta S)^2 + \Theta \Delta t + \mathcal{V}\Delta\sigma \tag{2.23}$$

2.5.2 Delta and Gamma Hedging Mechanics

- **Delta Neutrality:** A portfolio Π is delta-neutral if $\Delta_P = \frac{\partial \Pi}{\partial S} = 0$. For a portfolio containing an option position with total delta Δ_V, delta neutrality is achieved by shorting Δ_V units of the underlying asset:

$$\Pi = V - \Delta_V \cdot S \implies \Delta_\Pi = \Delta_V - \Delta_V = 0 \tag{2.24}$$

  Because Δ changes as S moves, delta hedging requires continuous dynamic rebalancing.

- **Gamma Neutrality:** Gamma measures the rate of change of Delta ($\Gamma = \frac{\mathrm{d}\Delta}{\mathrm{d}S}$). A high absolute gamma implies rapid delta slippage. Because the underlying asset has zero gamma ($\Gamma_{\text{asset}} \equiv 0$), gamma neutrality requires trading another option contract (V_T):

$$\Gamma_P + w_T \Gamma_T = 0 \implies w_T = -\frac{\Gamma_P}{\Gamma_T} \tag{2.25}$$

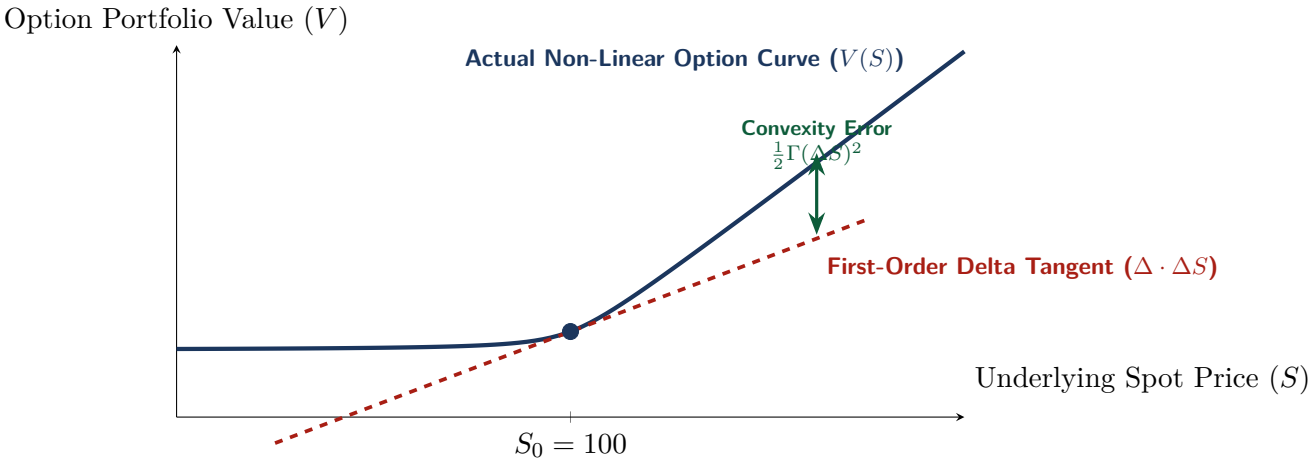


Figure 2.4: Non-Linear Option Price Curvature, First-Order Delta Tangency, and Gamma Error.

Once gamma is neutralized using the hedging option, the underlying asset is traded to restore delta neutrality.

Example 2.5.1: Delta-Gamma Hedging and Dynamic Portfolio Immunization

A market maker's option book is delta-neutral but possesses significant short gamma exposure:

- Portfolio Delta: $\Delta_P = 0$
- Portfolio Gamma: $\Gamma_P = -4,000$
- Traded Call Option 1 Gamma (Γ_1): $+1.60$
- Traded Call Option 1 Delta (Δ_1): $+0.55$

Task:

1. Determine the number of Option 1 contracts (w_1) needed to render the overall portfolio gamma-neutral.
2. Determine the quantity of underlying shares (ΔS) required to restore delta neutrality after the gamma hedge is executed.

Solution:

1. **Gamma Neutralization:**

$$\Gamma_{\text{new}} = \Gamma_P + w_1\Gamma_1 = 0 \implies -4,000 + w_1(1.60) = 0$$

$$w_1 = \frac{4,000}{1.60} = +2,500 \text{ Option 1 Contracts (Long)}$$

2. **Delta Restoration:** Purchasing $2,500$ contracts of Option 1 introduces positive delta into the book:

$$\Delta_{\text{introduced}} = w_1 \times \Delta_1 = 2,500 \times 0.55 = +1,375$$

Total portfolio delta is now $\Delta_{\text{current}} = 0 + 1,375 = +1,375$. To restore complete delta neutrality ($\Delta_{\text{target}} = 0$), the desk must sell $1,375$ units of the underlying stock:

$$\text{Stock Trade} = -1,375 \text{ shares (Short)}$$

The combined portfolio is now simultaneously Delta-neutral ($\Delta = 0$) and Gamma-neutral ($\Gamma = 0$).

Comprehensive Formula Reference Sheet

Chapter 1: Introduction to Risk Management

Log Return and Volatility Annualization

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right), \qquad \sigma_{\text{ann}} = \sigma_{\text{daily}} \times \sqrt{252} \tag{2.26}$$

Where:

- P_t = Asset price at time t
- P_{t-1} = Asset price at time $t-1$
- r_t = Continuously compounded log-return over period $t-1$ to t
- σ_{daily} = Sample standard deviation of daily log-returns
- σ_{ann} = Annualized return volatility assuming 252 trading days

Single Index Systematic and Unsystematic Risk Decomposition

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\epsilon_i}^2, \qquad \beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} = \rho_{im}\frac{\sigma_i}{\sigma_m} \tag{2.27}$$

Where:

- σ_i^2 = Total variance of asset i
- β_i = Systematic risk sensitivity coefficient (Beta) of asset i
- σ_m^2 = Variance of benchmark market index returns
- $\beta_i^2\sigma_m^2$ = Systematic (market-wide, non-diversifiable) risk component
- $\sigma_{\epsilon_i}^2$ = Unsystematic (idiosyncratic, diversifiable) residual variance
- ρ_{im} = Correlation coefficient between asset i and market m

Commodity Cost of Carry Forward Price

$$F_0 = S_0 e^{(r+u-y)T} \tag{2.28}$$

Where:

- F_0 = Theoretical forward / futures price at inception $t = 0$
- S_0 = Spot price of physical commodity at $t = 0$
- r = Continuously compounded risk-free rate per annum
- u = Continuous storage and insurance carrying cost per annum (percentage of spot)
- y = Continuous convenience yield per annum
- T = Time to maturity of the forward contract in years

Macaulay and Modified Duration

$$D = \frac{1}{P}\sum_{t=1}^{n}\frac{t \cdot C_t}{(1+y)^t}, \qquad D^* = \frac{D}{1+y} = -\frac{1}{P}\frac{\mathrm{d}P}{\mathrm{d}y} \tag{2.29}$$

Where:

- D = Macaulay duration in years
- D^* = Modified duration measuring percentage price sensitivity
- P = Current dirty price of the bond
- C_t = Cash flow (coupon or principal repayment) at time t
- y = Periodic yield to maturity (YTM)
- n = Total number of cash flow periods to maturity

Bond Convexity and Second-Order Price Approximation

$$C = \frac{1}{P(1+y)^2}\sum_{t=1}^{n}\frac{(t^2+t)C_t}{(1+y)^t}, \qquad \frac{\Delta P}{P} \approx -D^*\Delta y + \frac{1}{2}C(\Delta y)^2 \tag{2.30}$$

Where:

- C = Convexity measure of the bond in years^2
- ΔP = Estimated absolute change in bond price
- Δy = Parallel shift in the yield to maturity
- $-D^*\Delta y$ = First-order linear duration contribution
- $\frac{1}{2}C(\Delta y)^2$ = Second-order non-linear convexity correction

Dollar Value of an 01 (DV01 / PVBP)

$$\text{DV01} = D^* \times P \times 0.0001 \tag{2.31}$$

Where:

- DV01 = Monetary loss resulting from a 1 basis point (0.01%) increase in yield
- D^* = Modified duration of the bond or portfolio
- P = Total current market value of the fixed income position
- 0.0001 = Decimal equivalent of 1 basis point (1 bp)

Covered Interest Parity (CIP) Forward Exchange Rate

$$F = S \times \left[\frac{1 + r_d \cdot T}{1 + r_f \cdot T}\right] \tag{2.32}$$

Where:

- F = Theoretical forward exchange rate (Domestic currency per unit of Foreign currency)
- S = Spot exchange rate (Domestic per Foreign)
- r_d = Domestic annualized nominal risk-free interest rate
- r_f = Foreign annualized nominal risk-free interest rate
- T = Tenor of the forward contract in years

Chapter 2: Value at Risk (VaR) Methods and Hedging

Value at Risk and Square-Root-of-Time Scaling

$$\mathbb{P}(L > \text{VaR}_\alpha) = 1 - \alpha, \qquad \text{VaR}_{\alpha,T} = \text{VaR}_{\alpha,1} \times \sqrt{T} \tag{2.33}$$

Where:

- L = Portfolio loss variable over holding period T ($L = -\Delta\Pi$)
- VaR_α = Value at Risk at confidence level $\alpha \in (0, 1)$
- α = Target statistical confidence level (e.g., 0.95 or 0.99)
- $\text{VaR}_{\alpha,1}$ = 1-day Value at Risk
- $\text{VaR}_{\alpha,T}$ = Scaled T-day Value at Risk under i.i.d. return assumption
- T = Regulatory or holding horizon in days

Expected Shortfall (Conditional VaR)

$$\text{ES}_\alpha = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha] = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u \, \mathrm{d}u \tag{2.34}$$

Where:

- ES_α = Expected Shortfall (Conditional VaR) at confidence level α
- L = Portfolio loss variable
- VaR_α = Value at Risk cutoff threshold at confidence level α
- VaR_u = Quantile loss corresponding to probability level u

Delta-Normal Parametric Portfolio VaR

$$\sigma_P^2 = \mathbf{w}^T \mathbf{\Sigma} \mathbf{w}, \qquad \text{VaR}_\alpha = z_\alpha \cdot \sigma_P \cdot V_0 \tag{2.35}$$

Where:

- $\mathbf{w}$ = $n \times 1$ Vector of asset portfolio weights $(w_1, \ldots, w_n)^T$
- $\mathbf{\Sigma}$ = $n \times n$ Variance-covariance matrix of asset returns
- σ_P = Portfolio standard deviation (volatility) per period
- z_α = Standard normal critical quantile value (1.645 for 95%, 2.326 for 99%)
- V_0 = Current baseline market value of the portfolio

Two-Asset Portfolio Volatility

$$\sigma_P = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2} \tag{2.36}$$

Where:

- w_1, w_2 = Weights invested in Asset 1 and Asset 2 ($w_1 + w_2 = 1$)
- σ_1, σ_2 = Return standard deviations of Asset 1 and Asset 2
- ρ_{12} = Correlation coefficient between returns of Asset 1 and Asset 2
- σ_P = Diversified two-asset portfolio standard deviation

Optimal Minimum-Variance Hedge Ratio

$$h^* = \frac{\text{Cov}(\Delta S, \Delta F)}{\sigma_F^2} = \rho_{SF} \frac{\sigma_S}{\sigma_F} \tag{2.37}$$

Where:

- h^* = Optimal minimum-variance hedge ratio
- ΔS = Change in spot price of hedged asset over the holding period
- ΔF = Change in futures price of hedging instrument over the holding period
- σ_S = Standard deviation of spot price changes ΔS
- σ_F = Standard deviation of futures price changes ΔF
- ρ_{SF} = Correlation coefficient between ΔS and ΔF

Optimal Number of Hedging Futures Contracts

$$N^* = h^* \frac{Q_A}{Q_F}, \qquad N^* = (\beta_P - \beta^*) \frac{V_P}{V_F} \tag{2.38}$$

Where:

- N^* = Optimal number of futures contracts to trade
- Q_A = Total quantity / units of the spot asset being hedged
- Q_F = Contract size / units covered by one futures contract
- β_P = Current systematic beta of the equity portfolio
- β^* = Target post-hedge systematic beta ($\beta^* = 0$ for complete immunization)
- V_P = Current market value of the equity portfolio
- V_F = Market notional value of one index futures contract ($I \times M$)

Second-Order Taylor Series Greeks Price Approximation

$$\Delta V \approx \Delta \cdot \Delta S + \frac{1}{2}\Gamma(\Delta S)^2 + \Theta \Delta t + \mathcal{V} \Delta \sigma \tag{2.39}$$

Where:

- ΔV = Change in option value
- $\Delta = \frac{\partial V}{\partial S}$ = Option Delta (first-order price sensitivity)
- $\Gamma = \frac{\partial^2 V}{\partial S^2}$ = Option Gamma (curvature sensitivity)
- $\Theta = \frac{\partial V}{\partial t}$ = Option Theta (time decay sensitivity)
- $\mathcal{V} = \frac{\partial V}{\partial \sigma}$ = Option Vega (implied volatility sensitivity)
- ΔS = Change in underlying asset price
- Δt = Passage of time interval
- $\Delta \sigma$ = Change in implied volatility

Gamma Neutral Hedging Condition

$$w_T = -\frac{\Gamma_P}{\Gamma_T} \tag{2.40}$$

Where:

- w_T = Number of traded hedging option contracts to acquire or sell
- Γ_P = Total initial unhedged gamma of the primary option portfolio
- Γ_T = Gamma of one unit of the traded hedging option contract